

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Bundled Digital Financial Services for Smallholder Farmers

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 1 · Close the Climate Adaptation Gap through Scalable Climate Services and Smart Water Management

Flagship 1.1 · Digital Climate Services for Resilient Farm & Enterprise Management

Version 1.0 · June 2026



Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

This solution track addresses limited access to affordable financial services, a critical barrier to climate adaptation for smallholder farmers. Exclusion stems from three sources of market failure in rural credit markets: scarce collateral, the absence of complementary institutions such as functioning insurance markets, and information asymmetries between lenders and borrowers. Without assets to pledge, credit histories to present, or reliable mechanisms for lenders to assess their risk profiles, smallholder farmers – particularly women, youth, and tenant farmers – are either rationed out of formal lending entirely or face interest rates that reflect lender uncertainty rather than their actual creditworthiness. This prevents farmers from investing in climate-smart seeds, fertilizers, and other productivity-enhancing technologies.

The core innovation is a digital finance bundle co-developed and rigorously tested by CGIAR researchers with private sector partners. It is designed to address these inclusion barriers simultaneously. Remote sensing, climate, and soil data are used to determine a farmer's creditworthiness, addressing information asymmetries and the lack of physical collateral. Picture-based insurance (PBI), which uses smartphone images to assess crop damage and trigger payouts, protects both the farmer and the lender from production risks. By de-risking agricultural lending, the bundle unlocks capital for farmers to invest in climate-adaptive technologies.

This initiative aims to scale a proven bundled solution of digital credit and picture-based insurance to 100,000 farmers across Kenya and India by 2028. This scaling effort builds on over a decade of research and successful pilots, including randomized controlled trials (RCTs) with 5,000 farmers in India and Kenya. These trials validated the solution's impact, demonstrating increased access to formal credit, higher agricultural incomes, and improved women's empowerment and mental health. The track is led by two established social enterprises: **Dvara E-Registry** in India and **ACRE Africa** in Kenya, with technical and monitoring support from CGIAR centers (IFPRI, Alliance of Bioversity International and CIAT).

2. Innovation Bundle and Scaling Ambition

What is Being Scaled

- **Bundled Digital Credit and Insurance Product:** A financial service package combining a loan with comprehensive, picture-based crop insurance.
- **Digital Credit Scoring Technology:** Platforms (KhetScore in India, TARA in Kenya) that use remote sensing and climate analytics to assess creditworthiness, reducing the reliance on physical collateral to manage risk.
- **Picture-Based Insurance (PBI) Mechanism:** A claims settlement process that uses geo-tagged smartphone photos taken by farmers or agents to verify crop damage and trigger payouts, enhancing trust and transparency.
- **Capacity Building and Training Protocols:** Programs for financial institution staff and field agents on inclusive lending, digital tools, and the value proposition of bundled products.

Evidence of Impact

- **Increased Insurance Uptake:** In Kenya, using pictures to settle claims increased insurance uptake by **20 percentage points** and fertilizer use by **10 percentage points**.
- **Expanded Financial Inclusion:** In India, the credit-insurance bundle increased the use of formal credit by **20 percentage points** among small and marginal farmers.
- **Higher Incomes:** The same Indian pilot showed an increase in annual agricultural incomes of approximately **\$250 USD** per household.

- **Positive Wellbeing Impacts:** In India, farmers who were offered the bundle reported lower stress, easier loan repayment, and increased women's empowerment compared to farmers in the control group.

Scaling Targets

Outcome Metric	2026 Target	Long-term Goal (3–5 years)
Households reached	15,000 farmers (>12,000 India, >3,000 Kenya)	100,000 farmers (India and Kenya combined)
Inclusive reach	50% women; 25% youth (<35 years)	50% women; 25% youth (<35 years)
Area under improved practices	15,000 hectares	100,000 hectares
Investment leveraged (loans and insured value)	\$6,000,000 USD	\$25,000,000+ USD

3. Scaling Pathway and Theory of Change

Pathway Strategy The scaling pathway is a **private-sector-led, market-systems approach**. It leverages the existing operational infrastructure and commercial business models of two social enterprises, Dvara E-Registry and ACRE Africa. S4I funding acts as a catalytic accelerator, de-risking the expansion into new farmer segments and enhancing the product's technical foundation.

The strategy involves:

- **Strengthening the Core:** Using S4I funds to refine the underlying technology, specifically the computer vision models for loss assessment and integration of forward-looking climate risk analytics.
- **Building Capacity:** Training field agents and partner financial institution staff to ensure inclusive farmer onboarding and effective communication of the product's value.
- **Crowding-in Partners:** Leveraging the evidence generated from this scaled rollout to attract additional financial institutions and social investors, expanding the network of lenders and insurers offering the bundle.
- **Demonstrating Viability:** Proving the business case for serving previously excluded farmers (women, youth, those without land titles) to encourage systemic adoption by the financial sector.

Theory of Change If smallholder farmers, including women and youth, gain access to a bundled credit and insurance product that substitutes digital data for physical collateral, **and** the solution is delivered through the established networks of motivated private sector partners who are trained in inclusive practices, **then** financial institutions will update their perceptions of farmer creditworthiness and extend credit to previously excluded segments, and farmers will adopt the bundle farmers will adopt the bundle, increasing their access to formal finance and reducing their production risk. This will **lead to** increased investment in climate-smart agricultural technologies, **ultimately contributing to** higher, more resilient farm incomes, improved livelihoods, and a systemic shift in how financial institutions serve smallholder farmers.

Key Activities Funded by S4I

- **Technology Enhancement:** Refining computer vision models for faster, more accurate loss assessment and integrating high-resolution climate risk data into partner platforms (KhetScore, TARA).

- **Capacity Building:** Training field agents and financial institution staff on the use of digital tools, inclusive onboarding, and the benefits of the bundled solution.
- **Partnership and Governance:** Formalizing partnership agreements, convening stakeholder engagement meetings, and providing targeted, temporary smart subsidies and first-loss default guarantees to stimulate initial uptake.
- **Monitoring, Evaluation, Learning and Impact Assessment (MELIA):** Tracking progress against targets, documenting impact stories, and conducting research on agent behavior and farmer adoption/disadoption patterns.

4. Partners and Roles

Partner Type	Specific Partners and Quantities	Role in Scaling Pathway
Demand Partners (End-users and Aggregators)	Smallholder farmers in Kenya and India (target of 100,000). Aggregators: Farm Producer Organizations (FPOs) in India; Microfinance Institutions (MFIs) and banks (e.g., Juhudi, NCBA) in Kenya.	End-users of the financial bundle; aggregators provide the last-mile channel for onboarding farmers and facilitating transactions.
Scaling Partners (Implementation and Delivery)	Dvara E-Registry (India), ACRE Africa (Kenya). Financial institutions: Tata AIG, Avanti Finance (India); APA, Britam (Kenya).	Lead implementers managing the digital platforms, field operations, and partnerships with lenders and insurers. They underwrite loans (Dvara) or provide the risk scoring and insurance platform (ACRE).
Research and MELIA Partners	CGIAR: IFPRI (Berber Kramer, Shalini Roy), Alliance of Bioversity International and CIAT (Julian Ramirez-Villegas, Harold Achicanoy).	Provide technical backstopping, advance the computer vision and climate models, lead the MELIA strategy, ensure gender and social inclusion, and generate evidence for further scaling.

Funding Model: S4I catalytic funds cover technology enhancement, capacity building, MELIA, and temporary, targeted de-risking mechanisms (smart subsidies, first-loss guarantees), while the core lending and insurance operations are financed by farmers (through interest and premiums) and the capital from partner financial institutions and social investors.

5. Enabling Environment and Risks

Opportunities

- **Established Partners:** Dvara and ACRE Africa have existing commercial operations, partnerships with financial institutions, and a proven demand for their services.
- **Strong Evidence Base:** A decade of rigorous research, including RCTs, provides a compelling business case for investors and financial institutions.
- **Growing Demand for Climate Finance:** Increased global awareness of climate risk is pushing investors and insurers to seek innovative solutions for climate adaptation in agriculture.

- **Digital Adoption:** Rising smartphone penetration and the spread of mobile money (e.g., M-Pesa in Kenya) create a conducive environment for digital financial services.

Challenges and Mitigation

- **Climate Change Uncertainty:** Historical data is becoming less reliable for pricing risk. *Mitigation:* Partners are integrating forward-looking climate risk analytics into their scoring platforms (KhetScore, TARA) with support from CGIAR scientists.
- **Achieving Economies of Scale:** Insurance pools require large numbers for risk diversification and long-term sustainability. *Mitigation:* Bundling insurance with credit creates a strong demand pull, and the focus is on proving the model with tens of thousands of farmers to attract the larger investments needed to reach hundreds of thousands.
- **Lender and Farmer Trust:** Financial institutions are wary of lending to an untested segment, and farmers may distrust new digital products. *Mitigation:* Smart subsidies and first-loss default guarantees incentivize initial lender participation, while the transparency of picture-based insurance builds farmer trust. MELIA will explicitly track trust as a key outcome.
- **Inclusive Delivery:** Risk of excluding farmers without smartphones or those marginalized by agent bias. *Mitigation:* Agents are trained to onboard non-smartphone users. The project will study agent behavior to identify and mitigate biases, ensuring outreach to women, youth, and marginalized groups.

6. Learning and Adaptation

Monitoring Focus

- **Process Outcomes:** Effectiveness of agent training, speed of claims processing and payout, functionality of the technology platforms.
- **Behavior Change:** Adoption rates (loan disbursed + insurance active), disaggregated by gender, age, and social group; patterns of disadoption; changes in farmer trust in financial providers; agent decision-making and potential biases.
- **System Outcomes:** Willingness of new financial institutions to partner; leverage of additional public and private investment; shifts in partner business models towards serving excluded groups.

Adaptive Management Learning will be embedded in the project cycle through:

- **Quarterly Review Meetings:** With the S4I program team and implementing partners to review progress against milestones, discuss challenges, and adapt work plans.
- **Collaborative Research:** Close collaboration with S4I's Area of Work 5 (Learning) to conduct targeted studies (e.g., A/B testing of agent incentives) and translate findings into actionable recommendations.
- **Annual Reporting:** Capturing lessons learned and impact evidence to inform replication and attract new partners for expansion beyond 2028.

7. Next Steps and Timeline (2026)

Q1	Finalize and execute contracts with Dvara E-Registry and ACRE Africa. Conduct inception meetings. Begin integration of refined climate risk and computer vision models.
Q2	Conduct training programs for field agents and partner financial institution staff (focus on the upcoming season in both countries). Begin farmer onboarding and loan disbursement for the first season. Launch baseline MELIA data collection.
Q3	Support loan disbursement and insurance enrollment for the second season. Continue agent training and technology refinement. Publish an outcome story on the IFPRI-Dvara research partnership.
Q4	Monitor loan repayment and support any insurance claims settlement from the first season. Analyze MELIA data on year-one adoption and inclusion. Prepare for annual review with S4I. Publish an outcome story on the IFPRI-ACRE Africa partnership.

Communications and Knowledge: Key products will include outcome stories highlighting the research-to-scale pathway with Dvara and ACRE Africa, peer-reviewed publications, policy briefs, and presentations at agricultural economics conferences. Knowledge will be shared via the S4I program's communication channels and partner networks.

8. Contact

- **Berber Kramer (IFPRI-Kenya)** | b.kramer@cgiar.org
- **Julian Ramirez-Villegas (Alliance Bioversity-CIAT)** | j.r.villegas@cgiar.org